Table B-2: Example data for one day at five sites

Site	Day	Ground person silica concentration (mg/m³)	ln (ground person silica concentration)
1	1	0.005	-5.298
2	1	0.012	-4.423
3	1	0.010	-4.605
4	1	0.008	-4.828
5	1	0.006	-5.116
		$\bar{x}$ =mean(sites)=	-4.854
		S_x=standard deviation(sites)=	0.359
		df1=Degrees of freedom=n_{sites}-1	4

From equation (B-3), UCL2 = Term1 + sqrt (Term2 + Term3).
Where:

Term1= $\bar{x}+0.5\,s^2$ = -4.854 + 0.5 x 0.359^2 = -4.789.

Term2= $\left(s_x \times \frac{t(0.95,\ df1)}{\sqrt{n_{sites}}}\right)^2$ = $(0.359 \times 2.1318/\text{sqrt}(5))^2$ = 0.1169, where 2.1318 is the 95th

percentile of the t distribution with 4 degrees of freedom.

Term3= $0.5^2\ s_x^4\left(\frac{df1}{\text{chisq}(0.05,df1)}-1\right)^2$ =$0.25\times0.359^4\times(4/0.7107-1)^2$=0.0886, where 0.7107 is

the 5th percentile of the chi square distribution with 4 degrees of freedom.

Finally, UCL2 = (-4.789 + sqrt(0.1169+0.0886)) = -4.335 and $UCL_M = e^{UCL2}$ = exp(-4.335) = 0.0131 < 0.05 mg/m³.

Thus, at the 95% confidence level, it can be concluded that the exposures for the job category are less than the NIOSH REL of 0.05 mg/m³ for the data in this example. The corresponding upper confidence limit for the operator is 0.024 mg/m³, as shown in the example worksheet, to which the same conclusion applies. Thus, these conclusions apply to the population of sites from which the sample sites were drawn, with 95% confidence for each job category, or 90% overall confidence for both job categories (operator and ground person) for the example data.

Study Design Considerations

Important issues are the choice of strategy 1 or strategy 2, and, for either strategy, the number of sites to select. Given the cost associated with moving the milling machine to each site, using fewer sites may result in cost savings for the testing. Using too few sites, however, could result in low statistical power, which is the probability of correctly identifying an arithmetic mean below the occupational exposure limit. For the purposes of this document this is the probability that the 95% upper confidence limit is less than the NIOSH REL of 0.05 mg/m³. General guidelines

for choosing the sampling strategy were provided in Appendix B. More detailed calculations for determining sampling strategy are provided here.

Tables C-2, C-5, and C-6 in Appendix C give the power (estimated using statistical simulation) for various geometric means (GMs) and geometric standard deviations (GSDs). Table C-2 is for strategy 2, one day at each site; Table C-5 is for strategy 1, three days at each site. The GMs considered are 0.006, 0.01, and 0.025 mg/m^3 (all less than the NIOSH REL). The 0.006 mg/m^3 value is based on two past studies [NIOSH 2013a,b]; the 0.025 mg/m^3 value was added to see how large the GM can be and still have sizable power. The total GSDs from past studies varied between about 1.7 and 1.9 [NIOSH 2013a,b]; Table C-5 shows power results for total GSD 1.92, and Table C-6 shows results for a wider range of total GSDs, but only for GMs 0.006 and 0.01 mg/m^3.

The results in the Appendix C tables indicate that if the between-site variance is much larger than the within-site variance, little is gained in terms of power by sampling more than one day at each site. For example, for a GM of 0.006 mg/m^3 and a total GSD of 1.92, the power of strategy 2 (Table C-2) with 5 sites is 82.2% whereas the power of strategy 1 with 5 sites when most of the variability is between-sites is 84.7% (Table C-5). Thus strategy 2 is preferable because there is no real benefit to the extra work of collecting three days of data at each site. However, if the within-site variance is larger than the between-site variance, then sampling more than one day at each site will increase the study power (e.g., for a GM of 0.025 mg/m^3 and a total GSD of 1.92, the power of strategy 2 with 5 sites is 23.0% (Table C-2), but increasing the number of sampling days at each site from 1 to 3 increases the study power to around 48% when most of the variability is within-site (Table C-5)). Thus, strategy 1 would be preferred over strategy 2 for this example. This is a general conclusion of the effect of large versus small between-site variance that applies whenever the comparison between strategies is made for the same number of sites for the two strategies. The problem is that there is no way to know whether between or within- site variance is larger without collecting data, and, it is expensive to do a preliminary study to provide information about this question.

Based on the data available in two NIOSH reports [NIOSH 2013a,b] the recommendation is to use one of the following options:

Strategy 1: Use four sites, with three days per site. From Table C-6, for a total GSD of about 1.92, the power is between 66% and 94% for a GM of 0.006 mg/m^3. From two past NIOSH reports, 0.006 mg/m^3 is an attainable value [NIOSH 2013a,b]. If the GSD is about 2.25, the power is between 46% and 79% for GM 0.006 mg/m^3. For each interval, as explained above, the lower values (66% and 46%) are at the highest between-site variance shown.

Strategy 2: Use five sites, with one day per site. For GSD of about 1.9, the power is 82% and for GSD 2.25 the power is 61%, both for GM 0.006 mg/m^3 (Table C-2).

Both strategies offer reasonably high power (minimum close to 50% and maximum at least 80%) even if the total GSD is somewhat larger than that observed in past studies (GSD 1.92). Because of limited information about the between-site variance, there is no way to know beforehand which design is preferable. If the GM is 0.01 mg/m^3, for GSD 1.92, both strategies still have at least 50% power.

If the strategy 1 and 2 options above are impractical, there is an option with less power:

Strategy 1, 3 sites: Use three sites, with three days per site, with the understanding that although this design lacks power, it could still work under some conditions. The power is between 39% and 67% for a GSD of 1.92 and a GM of 0.006 mg/m^3 and between 28% and 49% for a GSD of 2.25 and GM of 0.006 mg/m^3 (Table C-6). The lower values are at larger between-site variance and the higher values are at lower between-site variance. Note that this is clearly inferior in power to the four site strategy 1 design results given above.

These recommendations are based on the observation that the total GSD from two past studies did not exceed 1.9 [NIOSH 2013a,b]. It is possible that future studies could result in larger GSDs that would require more sites to offer sufficient power, under either strategy, but this can only be determined when future results are obtained. If GMs are low, for example 0.01 mg/m^3 or less, but the upper confidence limits exceed the REL, this indicates lack of sufficient statistical power for the obtained GSDs. The power results in Tables C-2 and C-6 can help determine the number of sites required in a follow-up study to achieve adequate power for some larger GSDs. Note that it is imperative to decide beforehand on the sample size and not change the sampling plan after starting to collect and analyze data. Thus, if after collection of data it seems that more sites should have been sampled, a new study must be done using the revised number of sites.

Additional Comments

1. During data analysis, there may be situations where a sample value is missing (due to a sampling problem), unknown (perhaps because the sample determination is less than the limit of detection), or determined to be an outlier. When a sample is missing because of a sampling problem or excluded as an outlier the MOVER method can be used without the missing (or excluded) sample result. When a measurement is unknown because the determination is less than the limit of detection (LOD), a choice must be made among the following substitution procedures. It is not appropriate to exclude the unknown measurement in this case. If there is more than one value below the LOD, a statistician should be consulted; however, when there is just one such value, a simple method is to replace a single value below the LOD with the limit of detection divided by the square root of 2 [Hornung and Reed 1990].
2. The statistical methods proposed here work best if a) the same individual does the job at each site or b) different workers do the job at different sites. This applies for both the operator and ground person. If other arrangements are used, a statistician should be consulted.
3. The Excel worksheets in the spreadsheet "mover_upper_confidence_limits.xlsx" are provided so that the users of this document can enter their data into the appropriate sheet, and obtain 95% upper confidence limits for the arithmetic means of their data. See Appendix D for directions for using the spreadsheet.
4. The Silica/Asphalt Milling Machine Partnership recommends that asphalt milling machine manufacturers implement dust controls capable of reducing worker exposures below the NIOSH REL for respirable crystalline silica of 0.05 mg/m^3 at the 95% confidence level. Several asphalt milling machine manufacturers have also developed their dust controls to reduce worker exposures below the ACGIH TLV value for respirable crystalline silica of 0.025 mg/m^3. The upper confidence limits provided in this document and in the spreadsheet can be used for comparison to either occupational exposure limit.

5. In the methods presented here, each job category is evaluated separately, so that the results apply individually at the 95% confidence level and for both job categories with about 90% confidence.
6. Field data may include outliers. There are statistical methods for identifying outliers. Table B-3 contains 99% limits for the Grubbs statistic for a single upper outlier. For the natural log scale data, the calculated value

   g = [(maximum- mean)/ standard deviation]

   should be compared to the applicable Grubbs statistic, and if the value exceeds the 99% value in Table B-3, that data point could be removed. However, it is always important to have a reason based on some peculiarity of the particular sample to justify removing the data point. The formula can be used as follows:

   a) For either sampling strategy 1 or 2, the value of n should be the number of sites. For sampling strategy 1 the "maximum" is the largest site mean, the "mean" is the mean (site means), and the "standard deviation' is the standard deviation (site means). For sampling strategy 2 the "maximum" is the largest site measurement, the "mean" is the mean(sites) and the "standard deviation" is the standard deviation(sites). For example, from Table B-2, the maximum log scale site measurement is (-4.423) where n=5 and g=(-4.423-(-4.854))/0.359=1.201. From Table B-3 the 99th percentile of the Grubbs statistic for n=5 is 1.75. Because 1.201 < 1.75, there is no reason to consider the largest measurement to be an outlier.

   b) For sampling strategy 1, an additional use of the formula can be made for evaluation at a particular site where n=the number of days at the site=3. The "maximum" is the largest measurement for the site, the "mean" is the mean for the site, and the "standard deviation" is the standard deviation for the site. If multiple sites are tested, this application should only be performed in consultation with a statistician, because of complications that involve use of multiple tests.

   c) The g test can be performed with either the maximum or the minimum, whichever is considered a possible outlier. Use of the minimum will result in a negative value. To determine whether the minimum is a possible outlier, change the sign to positive and compare to the value in Table B-3 for the given n value.

Table B-3. 99% values for Grubbs test for a single upper outlier. [Barnett and Lewis 1978]

n*	99% value
3	1.15
4	1.49
5	1.75
6	1.94
7	2.10

* For sampling strategy 1, "n" = n_{sites} when determining if a site mean is an outlier compared to the other site means and "n" = n_{days} when determining if a within-site test result is an outlier compared to other site-specific results. For sampling strategy 2, "n" = n_{sites}.

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Appendix C: Explanation of Statistical Methods

The criterion used for evaluating the dust control strategy is that the arithmetic mean of the exposure measurements for a randomly chosen day at a randomly chosen site should be less than the occupational exposure limit (here the NIOSH REL of 0.05 mg/m^3 is used). This will be assessed by obtaining an upper 95% confidence limit for the arithmetic mean. If this upper 95% confidence limit is less than the occupational exposure limit then the exposures are said to be acceptably controlled.

Two sampling strategies are considered: 1) multiple sites (at least three), with three sampling days at each site and 2) multiple sites (at least five), with one sampling day at each site.

Under sampling strategy 1, for each of the two job categories the data are treated as coming from a one-way random effects model on the natural log scale:

$$x_{sd} = \mu + a_s + b_{d(s)}, \qquad \text{(C-1)}$$

where "s" indexes the sites, "d(s)" indexes the days at the sites, x_{sd} is the natural logarithm of the sampling result at site "s" on day "d", and μ is the mean on the natural log scale (i.e., the natural log of the geometric mean(GM)). Site is treated as a random effect (a_s) which is normally distributed with mean 0 and between-site variance σ^2_s (on the natural log scale). The days at each site are also treated as random (error) which is normally distributed with mean 0 and days within-site variance $\sigma^2_{d(s)}$ (on the natural log scale). The arithmetic mean is given by $\exp(\mu + 0.5(\sigma^2_s + \sigma^2_{d(s)}))$.

Under sampling strategy 2, the model simplifies to:

$$x_s = \mu + a_s, \qquad \text{(C-2)}$$

where "s" indexes the sites, x_s is the natural logarithm of the sampling result at site "s", and μ is the mean on the natural log scale (i.e., the natural log of the GM). Site is treated as random (error) which is normally distributed with mean 0 and between-site variance σ^2_s (on the natural log scale). In this scenario the arithmetic mean is given by $\exp(\mu + 0.5(\sigma^2_s))$.

In the following, methods are given for estimating the 95% upper confidence limit of the arithmetic mean. Simulation results will be provided a) to evaluate the coverage of the proposed methods for a wide range of possible parameter values and b) to give an idea of the power of the methods based on estimated means and variances from past data. Sampling strategy 2 is discussed first, because it is simpler.

Sampling Strategy 2

Here, the goal is to estimate a 95% upper confidence limit for the arithmetic mean, $\exp(\mu + 0.5(\sigma^2_s))$. There are exact confidence limits for the lognormal mean based on Land's coefficients [Gibbons and Coleman 2001] and [Land 1972], when the data are a simple random sample. However, this approach requires using tables that depend on the number of sites, thus requiring separate work sheets for each possible sample size. Instead, an approach is presented here that can be programmed in a single worksheet. The MOVER approach described by [Zou et. al. 2009] is

used to obtain an upper confidence limit for $\mu + 0.5(\sigma^2_s)$ and this result is exponentiated to obtain an upper confidence limit for the arithmetic mean.

According to [Zou et. al. 2009], for unbiased, statistically independent estimators $\hat{\theta}_1, \hat{\theta}_2,$ with 95% upper confidence limits u_1, u_2, an upper 95% confidence limit for the linear combination $a_1\theta_1 + a_2\theta_2$ is:

$$a_1\hat{\theta}_1 + a_2\hat{\theta}_2 + \sqrt{(a_1^2(u_1 - \hat{\theta}_1)^2 + a_2^2(u_2 - \hat{\theta}_2)^2)}$$

In this application, θ_1 is the natural log scale mean, θ_2 is the natural log scale variance, a_1=1 and a_2=0.5. In the formulation of [Zou et. al. 2009], the required upper confidence limit for the data described by Equation (C-2) is:

$$UCL2 = \bar{x} + 0.5s_x^2 + \sqrt{(s_x \times \frac{z(0.95)}{\sqrt{n_{sites}}})^2 + 0.5^2 s_x^4 (\frac{df\ 1}{chisq(0.05, df1)} - 1)^2} \quad \text{(C-3)},$$

where $\bar{x}$ is the natural log scale mean, s_x is the natural log scale standard deviation, z(0.95) is the 95th percentile of the standard normal distribution, df1 = n_{sites}-1, and chisq(0.05,df1) is the 5th percentile of the chi square distribution with df1 degrees of freedom.

(The derivation in [Zou et al. 2009] uses the proportionality of the squared differences $\left(u_i - \hat{\theta}_i\right)^2$ to the variance of $\hat{\theta}_i$ in large samples to justify the expression UCL2. When independence is lacking, a term for the covariance must be included.)

Because the sampling strategy will likely include a small number of sites, consider a modification that the z value be replaced by the analogous t-distribution value with df1 degrees of freedom, already shown as Equation (B-3):

$$UCL2 = \bar{x} + 0.5\, s_x^2 + \sqrt{(s_x \times \frac{t(0.95, df1)}{\sqrt{n_{sites}}})^2 + 0.5^2\, s_x^4 (\frac{df1}{chisq(0.05, df1)} - 1)^2}$$

where df1 = n_{sites}-1 and t(0.95,df1) is the 95th percentile of the t distribution with df1 degrees of freedom.

For either method, the MOVER upper confidence limit (UCL_M) is obtained by exponentiation of this result, as in Equation (B-4):

$UCL_M = e^{UCL2}$=exp(UCL2).

Statistical simulation was used to compare the coverage of these two methods:

1) Method 1 uses MOVER, as in equation (6) of [Zou et al. 2009] with the 95th percentile of the z (standard normal) distribution for the confidence interval of the mean.

2) Method 2 uses MOVER, but replaces the 95th percentile of the z (standard normal) distribution with the 95th percentile of the t distribution with k-1 degrees of freedom, where k is the number of sites (one measurement at each site).

For these simulations, the between-site variance (on the natural log scale) was varied from 0.25 to 5 and the population mean μ was set to $-0.5\sigma^2$, where σ^2 is the log scale variance. (Most of these values were used in [Zou et al. 2009]'s Table 1, although they calculated two-sided 95% confidence intervals, rather than the one-sided 95% upper confidence limits computed in these simulations). In this case, the upper confidence limit is for the arithmetic mean = $\exp(\mu+0.5\sigma^2)=\exp(0)=1$, or, equivalently, 0 on the natural log scale. By making the arithmetic means the same, the upper confidence limits can be compared for different variance values. The coverage is the percentage of simulations for which the arithmetic mean is less than the upper confidence limit. However, the coverage is not affected by the value of the arithmetic mean. For each combination of k (number of sites) and σ^2 (between-site variance), 50,000 simulations were performed. The results are described in Table C-1.

Table C-1: Coverage (expressed as percentage) for nominal 95% upper confidence intervals for (natural log scale) arithmetic mean of lognormal data from a random sample of 1 day at each of k sites, by 2 methods.

		k=5 sites		k=6 sites		k=7 sites	
GSD+	Log scale variance, σ_s^2	Method 1 MOVER, Using z statistic	Method 2 MOVER, Using t statistic	Method 1 MOVER, Using z statistic	Method 2 MOVER, Using t statistic	Method 1 MOVER, Using z statistic	Method 2 MOVER, Using t statistic
1.65	0.25	93.4 (0.69) *	95.0 (0.75)	93.0 (0.53)	94.7 (0.58)	93.1 (0.45)	94.6 (0.49)
2.03	0.50	93.7 (1.26)	94.8 (1.33)	93.8 (0.96)	94.8 (1.01)	93.7 (0.795)	94.8 (0.84)
2.72	1	94.3 (2.43)	95.0 (2.49)	94.2 (1.81)	95.0 (1.87)	94.1 (1.47)	94.9 (1.52)
3.40	1.5	94.5 (3.52)	95.2 (3.65)	94.4 (2.65)	95.0 (2.71)	94.4 (2.13)	95.0 (2.19)
4.11	2	94.7 (4.75)	95.2 (4.83)	94.6 (3.48)	95.1 (3.54)	94.6 (2.8)	95.0 (2.85)
4.86	2.5	94.7 (5.91)	95.2 (5.99)	94.9 (4.33)	95.3 (4.39)	94.6 (3.47)	95.0 (3.52)
5.65	3	95.0 (7.06)	95.3 (7.14)	94.9 (5.21)	95.2 (5.28)	94.6 (4.14)	95.0 (4.19)
9.36	5	95.0 (11.69)	95.2 (11.82)	95.0 (8.55)	95.2 (8.71)	94.8 (6.83)	95.0 (6.83)

* Cell entry is the actual coverage (the average value of the upper confidence limit on the log scale is in parentheses). For all simulations, the arithmetic mean converted to the log scale is 0 and there are $n_{days}=1$ at each site. Results based on 50,000 simulations.

+ GSD= $e^{\sqrt{\sigma_s^2}}$

The results show that method 1 using the z percentiles gives coverage less than 95%, especially for the smaller variances. This is consistent with [Zou et al. 2009]'s Table 1, where, for the smaller variances, the coverage of the upper confidence limit for method 1 is less than the nominal value.

The use of the t percentiles in method 2 corrects for this. The average value of the upper confidence limits is somewhat larger for method 2, since that method has larger coverage.

Statistical simulation was also used to calculate the power for the MOVER method based on the t-statistic. For the simulations done here the GM of 0.006 mg/m^3 was used because it was the overall GM of geometric means from two past studies [NIOSH 2013a,b]. In addition, computations for GMs of 0.01 and 0.025 mg/m^3 have been included in the table. Total geometric standard deviation (GSD) for the data from past studies varied between about 1.7 and 1.9. Values of this magnitude as well as those somewhat smaller and somewhat larger were used in the calculations, as shown in the table. If future studies have approximately the specified GMs and GSDs in the range shown in the table, then the probabilities of meeting the REL will be approximately those shown in Table C-2. Coverage is the fraction of simulations such that [ln(GM) + 0.5 σ_s^2] is less than the 95% upper confidence limit for the MOVER method 2, using the t statistic (Equation (B-3)). Power is the fraction of simulations such that the upper confidence limit for MOVER method 2 will be less than the NIOSH REL of 0.05 mg/m^3.

Table C-2: Coverage and power (expressed as percentages) for strategy 2 data, using Mover method 2.*

GM+	GSD	Log scale variance, σ_s^2	5 sites		6 sites		7 sites	
			Coverage	Power	Coverage	Power	Coverage	Power
0.006	1.4	0.11	94.8	100.0	94.6	100	94.5	100.0
	1.75	0.31	94.9	92.5	94.7	98.1	94.9	99.6
	1.92	0.426	94.9	82.2	94.8	92.4	94.6	97.3
	2.25	0.66	95.0	61.1	95.1	74.4	94.7	84.1
	3.0	1.21	95.4	31.4	95.1	39.7	94.9	47.9
0.01	1.4	0.11	94.5	99.9	94.4	100.0	94.6	100.0
	1.75	0.31	94.9	82.1	94.8	92.0	94.6	97.0
	1.92	0.426	95.0	67.6	94.8	81.0	94.7	89.0
	2.25	0.66	95.0	45.1	95.1	56.0	94.9	66.0
	3.0	1.21	95.2	20.6	95.1	26.0	95.3	31.0
0.025	1.4	0.11	94.6	81.0	94.4	91.0	94.7	96.0
	1.75	0.31	94.8	34.0	94.9	42.0	94.4	50.0
	1.92	0.426	94.9	23.0	94.7	28.0	94.8	34.0
	2.25	0.66	95.0	13.0	95.0	15.0	94.7	18.0
	3.0	1.21	95.1	5.8	95.0	6.2	95.1	6.3

* For all simulations, there are n_{days}=1 at each site. Results based on 50,000 simulations.

The coverage is slightly anti-conservative for the variances shown here, which is consistent with Table C-1 results for method 2 for relatively small variances. In general, the coverage is quite close to 95% for method 2; coverage is never less than 94.4% and never more than 95.4%.

For larger variances, power is much larger when 7 sites are used. However, for the GSD magnitude similar to that in previous studies (1.92), the chances of meeting the REL are quite high (>80%) for 5 sites if the GM is as low as 0.006 mg/m^3. For the same GSD and GMs of 0.01 and 0.025 mg/m^3, respectively, the power drops to 68% and 23%, respectively.

Sampling Strategy 1 – Balanced Design

The goal is to estimate a 95% upper confidence limit for the arithmetic mean, $\exp(\mu + 0.5(\sigma^2_s + \sigma^2_{d(s)}))$, for the data model given by Equation (C-1). Sampling strategy 1 data are not a simple random sample since there are multiple days at each site, so the statistical methods are more complicated. For large samples, the delta method approximation can be used in conjunction with maximum likelihood or ANOVA estimation to calculate confidence levels [Mandel 2013], but the sample sizes here are all small. Therefore, the choice of nominal confidence level for these methods must be calibrated by simulation, because nominal confidence differs from actual confidence. There is an approach based on generalized confidence intervals that can be used, although for small samples it must also be calibrated by simulation [Krishnamoorthy and Mathew 2002]. The work of [Zou et al. 2009], however, can be easily generalized to the case of two variance components, as in [Krishnamoorthy and Peng 2014], equation (2), as follows.

For statistically independent estimators , $\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3$, with 95% upper confidence limits u_1, u_2, and u_3, and 5% lower confidence limits l_1, l_2, and l_3, an upper 95% confidence limit for a linear combination is $a_1\theta_1 + a_2\theta_2 + a_3\theta_3$:

$$a_1\hat{\theta}_1 + a_2\hat{\theta}_2 + a_3\hat{\theta}_3 + \sqrt{a_1^2\left(u_1^* - \hat{\theta}_1\right)^2 + a_2^2\left(u_2^* - \hat{\theta}_2\right)^2 + a_3^2\left(u_3^* - \hat{\theta}_3\right)^2}$$

where $u_i^* = u_i$ if $a_i > 0, l_i$ if $a_i < 0$.

Interest is in estimation of the upper confidence limit for the linear combination

$\mu + 0.5\ (\sigma^2_s + \sigma^2_{d(s)})$

$= \mu + 0.5\ [(\sigma^2_s + \sigma^2_{d(s)}\frac{1}{n_{days}}) + \frac{n_{days} - 1}{n_{days}}\sigma^2_{d(s)}]$

In this application, θ_1 *is* μ (natural log of the GM); for n_{days} at each site, θ_2 is the variance of site means for natural log transformed data ($\sigma^2_s + \sigma^2_{d(s)}\frac{1}{n_{days}}$); and θ_3 is the within-site variance ($\sigma^2_{d(s)}$). Furthermore, all a_i>0 (specifically, a_1=1, a_2=0.5, and for balanced data with n_{days} at each site, a_3=0.5 x (1-1/n_{days})). However, contrary to the assignment rule in [Krishnamoorthy and Peng 2014], simulation results indicate that for the sample sizes of interest in our study, coverage is much closer to nominal when $u_3^* = l_3$. [Krishnamoorthy 2014] recommends this alteration and simulation results are shown below that also support this choice.

Thus, the required upper confidence limits can be obtained as follows:

The sample variance of the natural log scale site means ($s^2_{\bar{x}_s}$ in equation B-1) estimates θ_2 and the average within-site variance (denoted $astd^2_{ws}$) estimates $\theta_3 = \sigma^2_{d(s)}$. Thus, $\hat{\theta}_1 = \bar{\bar{x}}$ the average of the log scale site means, $\hat{\theta}_2 = s^2_{\bar{x}_s}$, and $\hat{\theta}_3 = astd^2_{ws}$. Upper confidence limits for the corresponding parameters are given by $u_1 = \bar{\bar{x}} + s_{\bar{x}_s} \times t_{(0.95,\, df1)} / sqrt(n_{sites})$, $u_2 = df1 \times s^2_{\bar{x}_s} / chisq(0.05,df1)$, and $l_3 = df2 \times astd^2_{ws} / chisq(1-0.05,df2)$, respectively, where $astd_{ws}$ is the average standard deviation within sites (square root of the average of the squares of the within-site standard deviations for balanced data), df1 = n_{sites}-1, and (for balanced data) df2 = n_{sites} x (n_{days}-1). Also, t(0.95,df1) denotes the 95th percentile of the t distribution with df1 degrees of freedom; chisq(0.05,df1) denotes the 5th percentile of the chi square distribution with df1 degrees of freedom; and chisq(1-0.05,df2) denotes the 95th percentile of the chi square distribution with df2 degrees of freedom. Making substitutions and simplifying yields the required upper confidence limit for balanced data, shown in Equation (B-1):

$$UCL2 = \bar{\bar{x}} + 0.5 s^2_{\bar{x}_s} + 0.5 \frac{(n_{days} - 1)}{n_{days}} astd^2_{ws} + \sqrt{\left(s_{\bar{x}_s} \frac{t(0.95, df1)}{\sqrt{n_{sites}}} \right)^2 + 0.5^2 s^4_{\bar{x}_s} \left(\frac{df\,1}{chisq(0.05,df1)} - 1 \right)^2 + \left(0.5 \frac{n_{days} - 1}{n_{days}} \right)^2 astd^4_{ws} \left(\frac{df\,2}{chisq(1-0.05,df2)} - 1 \right)^2}$$

$UCL_M = e^{UCL2}$ is the required upper confidence limit, given above in Equation (B-2).

(Equation (B-1) is analogous to Equation (6) of [Zou et. al. 2009].)

Statistical simulation was used to compare the coverage of the two methods:

1) MOVER (using all upper confidence limits), denoted $UCL_{M,U}$
2) Modified MOVER (using upper confidence limits, except using the lower confidence limit for the variance between days), denoted UCL_M.

In these simulations, coverage was estimated for several balanced designs (3, 4, or 5 sites) based on 3 days per site. As in Table C-1 the aim is to make the upper confidence limits comparable. Analogous to Table C-1 the value of μ is $[-0.5 \times (\sigma_s^2 + \sigma^2_{d(s)})]$, where σ_s^2 is the between-site variance and $\sigma^2_{d(s)}$ is the days within-site variance. For each choice of variances, the upper confidence limit is for $\exp[\mu + 0.5 \times (\sigma_s^2 + \sigma^2_{d(s)})] = \exp(0) = 1$, or, equivalently, 0 on the natural log scale. In Table C-3 seven scenarios for σ^2_s and $\sigma^2_{d(s)}$ were considered, for $\sigma^2_s + \sigma^2_{d(s)} = 1$ in each scenario. The coverage is evaluated for a very wide range of the ratio of the between-site variance to the total variance, to demonstrate that the coverage is maintained, whatever that ratio is. For each set of parameter values, 50,000 simulations were performed. The coverage is the percentage of simulations for which the arithmetic mean is less than the upper confidence limit. Because the coverage will vary also with the magnitude of the total variance ($\sigma_s^2 + \sigma^2_{d(s)}$), Table C-4 presents simulations results for other total variance values, using four other variances from Table C-1, just for n_{sites} equal 3 and 5.

Table C-3: Coverage (expressed as percentage) for nominal 95% upper confidence intervals for (natural log scale) arithmetic mean of lognormal data for strategy 1 for the two MOVER methods when log scale total variance is 1, and ln (arithmetic mean)=0

			3 sites		4 sites		5 sites	
σ^2_s	$\sigma^2_{d(s)}$	$\sigma^2_s/(\sigma^2_s+\sigma^2_{d(s)})$	$UCL_{M,U}$	UCL_M	$UCL_{M,U}$	UCL_M	$UCL_{M,U}$	UCL_M
0	1	0	98.2 (3.49) *	95.5 (3.27)	97.8 (1.63)	95.0 (1.43)	97.4 (1.12)	94.5 (0.96)
0.2	0.8	0.2	97.9 (4.62)	95.5 (4.47)	97.3 (2.03)	95.1 (1.92)	96.7 (1.34)	94.8 (1.26)
0.3	0.7	0.3	97.6 (5.20)	95.5 (5.12)	96.8 (2.28)	95.3 (2.19)	96.5 (1.48)	95.1 (1.42)
0.5	0.5	0.5	96.9 (6.38)	95.4 (6.35)	96.2 (2.71)	95.3 (2.66)	95.9 (1.76)	95.1 (1.73)
0.7	0.3	0.7	96.0 (7.52)	95.5 (7.48)	95.5 (3.25)	95.3 (3.19)	95.6 (2.05)	95.1 (2.03)
0.8	0.2	0.8	95.8 (8.12)	95.5 (8.13)	95.5 (3.45)	95.2 (3.44)	95.4 (2.19)	95.1 (2.18)
0.95	0.05	0.95	95.4 (9.01)	95.6 (9.08)	95.3 (3.8)	95.3 (3.81)	95.1 (2.43)	95.2 (2.41)

*Cell entry is the actual coverage (the average value of the upper confidence limit on the log scale is in parentheses). Results based on 50,000 simulations. For all simulations, the natural log of the arithmetic mean is 0 in each simulation and there are ndays=3 at each site.

Table C-4 Coverage (expressed as percentage) for nominal 95% confidence limits and average upper confidence limits for the two MOVER methods for strategy 1 when natural log scale total variance equals 0.25, 2, 3, or 5, values that were used in Table C-1, and ln(arithmetic mean)=0.

	Total variances: (σ2s+σ2d(s))															
σ^2_s/ Total vari-ance	0.25				2				3				5			
	3 sites		5 sites		3 sites		5 sites		3 sites		5 sites		3 sites		5 sites	
	Method $UCL_{M,U}$	Method UCL_M	Method $UCL_{M,U}$	Method UCL_M	Method $UCL_{M,U}$	Method UCL_M	Method $UCL_{M,U}$	Method UCL_M	Method $UCL_{M,U}$	Method UCL_M	Method $UCL_{M,U}$	Method UCL_M	Method $UCL_{M,U}$	Method UCL_M	Method $UCL_{M,U}$	Method UCL_M
0	97.2 (0.95)*	95.1 (0.90)	96.3 (0.36)	94.6 (0.33)	98.6 (6.8)	95.6 (6.4)	97.8 (2.1)	94.4 (1.8)	98.7 (10.2)	95.5 (9.5)	98.0 (3.1)	94.6 (2.6)	98.8 (17.1)	95.6 (15.8)	98.1 (5.1)	94.7 (4.2)
0.2	96.6 (1.24)	95.1 (1.22)	95.6 (0.43)	94.4 (0.41)	98.1 (9.1)	95.6 (8.8)	97.4 (2.6)	95.0 (2.4)	98.4 (13.5)	95.8 (13.2)	97.6 (3.8)	95.3 (3.5)	98.5 (22.6)	95.9 (21.9)	97.8 (6.2)	95.2 (5.6)
0.3	96.4 (1.39)	95.0 (1.37)	95.5 (0.47)	94.8 (0.46)	97.9 (10.3)	95.8 (10.0)	97.0 (2.8)	95.1 (2.7)	98.2 (15.4)	95.6 (14.9)	97.4 (4.1)	95.2 (3.9)	98.3 (25.5)	95.6 (24.9)	97.5 (6.8)	95.2 (6.4)
0.5	95.8 (1.68)	95.1 (1.67)	95.2 (0.55)	94.7 (0.54)	97.2 (12.7)	95.6 (12.5)	96.3 (3.4)	95.0 (3.3)	97.6 (18.8)	95.7 (18.8)	96.5 (4.9)	95.1 (4.8)	97.6 (31.3)	95.4 (30.7)	96.8 (8.1)	95.1 (7.9)
0.7	95.6 (1.99)	95.2 (1.99)	95.0 (0.62)	94.7 (0.62)	96.5 (15.1)	95.6 (15.0)	95.7 (3.9)	95.3 (3.9)	96.4 (22.5)	95.5 (22.3)	95.8 (5.8)	95.0 (5.7)	96.6 (37.1)	95.4 (37.4)	95.5 (9.4)	95.1 (9.4)
0.8	95.5 (2.15)	95.5 (2.14)	94.7 (0.66)	94.6 (0.66)	95.8 (16.3)	95.5 (16.2)	95.3 (4.2)	95.3 (4.2)	95.8 (24.4)	95.4 (24.3)	95.6 (6.2)	95.2 (6.2)	95.9 (40.2)	95.4 (40.4)	95.3 (10.3)	95.2 (10.3)
0.95	95.3 (2.37)	95.1 (2.38)	94.7 (0.72)	94.9 (0.72)	95.5 (18.1)	95.4 (18.1)	95.0 (4.7)	95.2 (4.7)	95.3 (26.9)	95.5 (26.9)	95.2 (6.9)	95.1 (6.9)	95.2 (44.9)	95.0 (44.7)	95.2 (11.4)	95.0 (11.4)

* Cell entry is the actual coverage (the average value of the upper confidence limit on the natural log scale is in parentheses). Results based on 50,000 simulations. There are n_{days}=3 at each site.

In Tables C-3 and C-4, the upper confidence limits for the 5 site results for the ratio $\sigma^2_s/(\sigma^2_s+\sigma^2_{d(s)})=0.95$, should approximate those from Table C-1, for the corresponding total variances. All of these tables are for arithmetic mean 1 (0 on log scale.) It makes sense to use results for this variance from Table C-4 because for this situation the total variance is most like that in Table C-2 with one dominant component. They do agree approximately. For instance, for total variance 0.25, Table C-4 gives an upper confidence limit for both methods of about 0.72, compared with 0.75 for method 2 in Table C-1. Coverage is about 95% from both these tables.

The coverage results in Tables C-3 and C-4, for variances between 0.25 and 5 indicate that the modified version of the MOVER (method 2, UCL_M) gives coverage much closer to the nominal 95% than the original version (method 1, $UCL_{M,U}$). The modified MOVER (method 2) is slightly anti-conservative for 5 sites and total variance 0.25. The coverages for the method 2 are never lower than 94.4% and never higher than 95.9%.

Statistical simulation was also used to evaluate the power of the modified MOVER for parameter values in the range from past studies, analogous to results in Table C-2. The results are shown in Tables C-5 and C-6. All calculations use $n_{days}=3$ at each site. The GMs considered in Table C-5 are 0.006, 0.01, and 0.025 mg/m^3 (all less than the NIOSH REL). The GSD 1.92 is used because for several previous studies the total GSDs were between about 1.75 and 1.9. For instance, for $\sigma^2_s=0.106$ and $\sigma^2_{d(s)}=0.319$ (values from Table C-5) the total GSD is $\exp([0.106+0.319]^{0.5})\sim1.92$. The 0.025 mg/m^3 GM was added to see how large the GM can be and still have sizable power. In addition, in order to study the power further, Table C-6 gives additional power results for five other total GSDs used in Table C-2, but only for GMs of 0.006 and 0.01 mg/m^3. For the Tables C-5 and C-6, for each total GSD, the between- site variances shown are about 10%, 25%, 75% and 90% of the total variance.

Coverage is the fraction of simulations such that $[\ln(GM) + 0.5\,(\sigma^2_s+\sigma^2_{d(s)})]$ is less than the 95% upper confidence limit. Power is the fraction of simulations such that the upper confidence limit UCL_M is less than the NIOSH REL of 0.05 mg/m^3.

Table C-5: Coverage and power (expressed as percentages) for modified MOVER*

GM	Total GSD+	Between-site σ_s^2	Within-site $\sigma_{d(s)}^2$	3 sites		4 sites		5 sites	
				Coverage	Power	Coverage	Power	Coverage	Power
0.006	1.92	0.0426	0.383	95.3	67.1	94.9	93.8	94.8	99.5
0.006	1.92	0.106	0.319	95.4	59.2	94.9	88.4	94.8	98.2
0.006	1.92	0.319	0.106	95.4	42.5	95.2	71.3	94.9	88.2
0.006	1.92	0.383	0.0426	95.4	39.0	95.0	66.4	95.0	84.7
0.01	1.92	0.0426	0.383	95.3	54.8	95.0	84.6	94.6	96.4
0.01	1.92	0.106	0.319	95.4	47.6	94.9	77.0	94.8	92.4
0.01	1.92	0.319	0.106	95.4	33.4	95.1	56.3	94.8	75.1
0.01	1.92	0.383	0.0426	95.4	30.7	95.1	51.8	94.9	70.3
0.025	1.92	0.0426	0.383	95.4	22.6	94.9	35.0	94.7	47.9
0.025	1.92	0.106	0.319	95.4	19.4	95.0	30.2	94.7	40.7
0.025	1.92	0.319	0.106	95.4	14.0	95.1	20.6	94.8	27.0
0.025	1.92	0.383	0.0426	95.3	13.1	95.1	18.5	95.0	24.4

* For all simulations, there are n_{days}=3 days at each site. Results are based on 50,000 simulations.

+ Total GSD= $e^{\sqrt{\sigma_s^2+\sigma_{d(s)}^2}}$

Table C-6: Coverage and power (expressed as percentages) for strategy 1 data. Simulation results are shown for GM=0.006 and 0.01 mg/m³, and for all total GSDs used in Table C-2 ; 50,000 simulations for each set of parameters

				GM=0.006						GM=0.01					
				3 sites		4 sites		5 sites		3 sites		4 sites		5 sites	
Total GSD[+]	Approximate Ratio σ_s^2 /Total Variance	Between-site σ_s^2	Within-site $\sigma_{d(s)}^2$	Coverage	Power	Coverage	Power	Coverage	Power	Coverage	Power	Coverage	Power	Coverage	Power
1.4	0.10	0.011	0.099	95.1	98.9	94.6	100.0	94.5	100.0	94.9	96.3	94.6	100.0	94.5	100.0
1.4	0.25	0.028	0.083	94.9	97.3	94.6	100.0	94.7	100.0	94.7	93.0	94.7	99.9	94.6	100.0
1.4	0.75	0.083	0.028	94.8	88.6	94.8	99.8	94.7	100.0	95.1	80.1	94.9	98.7	94.8	100.0
1.4	0.90	0.099	0.011	95.2	86.0	94.9	99.5	94.6	100.0	95.2	76.1	94.7	97.8	94.6	99.9
1.75	0.10	0.03	0.28	95.3	78.5	94.6	98.4	94.7	100.0	95.3	67.4	94.9	94.0	94.7	99.4
1.75	0.25	0.078	0.23	95.3	71.3	95.0	95.9	94.6	99.7	95.2	59.9	94.8	88.9	94.7	98.2
1.75	0.75	0.23	0.078	95.5	53.4	95.1	83.8	95.1	96.4	95.3	43.1	95.2	71.4	94.9	88.5
1.75	0.90	0.28	0.03	95.4	49.3	95.2	79.7	94.7	94.4	95.3	39.9	95.0	66.3	94.8	84.8
1.92	0.10	0.0426	0.383	95.3	67.1	94.9	93.8	94.8	99.5	95.3	54.8	95.0	84.6	94.6	96.4
1.92	0.25	0.106	0.319	95.4	59.2	94.9	88.4	94.8	98.2	95.4	47.6	94.9	77.0	94.8	92.4
1.92	0.75	0.319	0.106	95.4	42.5	95.2	71.3	94.9	88.2	95.4	33.4	95.1	56.3	94.8	75.1
1.92	0.90	0.383	0.0426	95.4	39.0	95.0	66.4	95.0	84.7	95.4	30.7	95.1	51.8	94.9	70.3
2.25	0.10	0.066	0.59	95.5	49.3	94.8	79.4	94.7	93.8	95.4	38.3	95.0	63.6	94.8	82.5
2.25	0.25	0.166	0.495	95.5	42.7	95.0	70.5	94.9	88.5	95.6	32.5	95.2	54.6	94.9	73.5
2.25	0.75	0.495	0.166	95.4	30.2	95.1	50.5	95.0	68.8	95.2	23.1	95.0	37.0	94.9	52.1
2.25	0.90	0.59	0.066	95.3	28.1	95.3	46.2	95.0	64.2	95.6	21.4	95.1	34.2	95.1	47.2
3.0	0.10	0.12	1.09	95.8	28.5	95.1	47.7	95.0	64.6	95.4	19.9	95.2	32.0	94.8	44.5
3.0	0.25	0.30	0.91	95.5	24.2	95.4	39.9	95.2	55.5	95.6	17.6	95.3	27.2	95.0	37.1
3.0	0.75	0.91	0.30	95.6	17.2	95.3	26.5	95.3	36.9	95.5	12.6	95.2	18.5	95.3	23.9
3.0	0.90	1.09	0.12	96.0	15.8	95.2	24.5	95.1	33.3	95.3	12.2	95.4	16.8	95.2	21.7

[+] Total GSD=$e^{\sqrt{\sigma_s^2+\sigma_{d(s)}^2}}$.

Conclusions concerning coverage are that it is between 94.5% and 96.0%.

Conclusions concerning power from these two tables are:

1) It is useful to compare the power for five sites in Table C-5 (or C-6) to that in Table C-2 for GSD=1.92 and GM=0.006 mg/m^3. For Table C-2, the power is about 82%, compared to about 85% from Table C-5 for σ_s^2=0.383 and $\sigma_{d(s)}^2$ =0.0426. When all the variance in the two component model is due to between-site variance, this is the same as the one component model, since the degrees of freedom for sites are the same. Likewise, for Table C-2, for five sites and GSD=3.0 power is about 31.4%, which is close to 33.3% from Table C-6 for σ_s^2=1.09 and GM=0.006 mg/m^3. For both Tables C-5 and C-6, for each sample size and total GSD, the power decreases as the between-site variance becomes a larger portion of the total variance.

2) For the two lowest GMs shown (0.006 and 0.01 mg/m^3) in Table C-5 or C-6, GSD=1.92, the highest power for the 3 site data (67.1% for GM=0.006 mg/m^3 and 54.8% for GM=0.01 mg/m^3) is approximately the same as the lowest power for the 4 site data (66.4% for GM=0.006 mg/m^3 and 51.8% for GM=0.01 mg/m^3). Thus, there is a big difference in power between sampling at 3 or at 4 sites for total GSD 1.92. Examination of Table C-6 indicates similar results for other total GSDs, especially at the 0.006 mg/m^3 level.

3) If the GSD becomes much larger than 1.9, the power decreases considerably. This happens for both strategy 1 and strategy 2 designs. For the seven site strategy 2 design and GSD of 3.0 and GM of 0.006 mg/m^3, the power is less than 50% (Table C-2). For the 5 site strategy 1 design with total GSD 3.0 and between variance accounting for at least 75% of the total variance and GM 0.006 mg/m^3, the power is less than 40% (Table C-6). Thus, the discussion of power in this document relies considerably on observing total GSD much less than 3.

4) When the GM is 0.025 mg/m^3, in Table C-5, for no sample size (≤5 sites) is the power greater than 50%.

Sampling Strategy 1 – Unbalanced Design

Although the recommendation for strategy 1 is to sample three days at each site, there will probably be studies where that is not possible. For such unbalanced data, the same formula for the UCL can be used; however, the following modifications must be made:

1. df2 = number of measurements-number of sites.

2. astd_{ws}^2 is the sum of the weighted within-site variances divided by df2. Each weight is the number of days at the site minus 1.

3. In equation (B-1), $\frac{n_{days-1}}{n_{days}}$ is replaced by (1-nbar), where nbar is the average of the reciprocals of n_{days} the number of days at the sites. Note that in the case of

balanced data with n_{days} at each site, nbar is equal to $1/n_{days}$ and (1-nbar) reverts back to $\frac{n_{days-1}}{n_{days}}$.

4. $\sqrt{s^2_{\bar{x}_s} - nbar \times astd^2_{ws}}$ =standard deviation between sites; note that for balanced data, nbar=$1/n_{days}$.

For unbalanced data, $\bar{\bar{x}}$, the mean of site means on the natural log scale, is not statistically independent of $s^2_{\bar{x}_s}$ and simulations are done to allow for this. Also, $astd^2_{ws}$ is statistically independent of both $\bar{\bar{x}}$ and $s^2_{\bar{x}_s}$ and is generated that way in the simulations. Although there is some correlation of the estimators, Equation B-1 does not include any term for it. The simulations indicate that the coverage is acceptable. [Krishnamoorthy 2014] also does not include a term for the correlation of $\bar{\bar{x}}$ and $s_{\bar{x}_s}$. Note that $\bar{\bar{x}}$ and $s_{\bar{x}_s}$ still denote the natural log scale mean of site means and the standard deviation of site means, respectively.

Statistical simulation was used to estimate coverage for several unbalanced designs based on 3 sites and (up to) 3 days per site. For these simulations, the population mean μ was set to $-0.5(\sigma^2_s + \sigma_{(s)})$. Thus, for each choice of variances, the upper confidence limit is for $\exp[\mu+0.5 \times(\sigma_s^2 + \sigma^2_{d(s)})]=\exp(0)=1$, or, equivalently, 0 on the natural log scale, as was also done in Tables C-3 and C-4. By making the arithmetic means the same, the upper confidence limits can be compared for different variance values. For each of the variance settings, 50,000 simulations were performed. The coverage is the percentage of simulations for which the arithmetic mean is less than the upper confidence limit. Only the modified MOVER method, UCL_M, results are shown in Table C-7.

Table C-7: Coverage (expressed as percentage of simulations) of the modified MOVER method for the unbalanced design based on 3 and 5 sites.

		Number of days at each site with k=3 sites:				Number of days at each site with k=5 sites:			
σ^2_s	$\sigma^2_{d(s)}$	(3,3,3)	(3,3,2)	(3,2,2)	(3,3,1)	(3,3,3,3,3)	(3,3,3,3,2)	(3,3,3,2,2)	(3,3,2,2,1)
0	1	95.5(3.27)*	95.5 (3.77)	95.5(4.26)	95.5(5.27)	94.5(0.96)	94.4(1.03)	94.6(1.11)	94.6(1.41)
0.2	0.8	95.5 (4.47)	95.5 (4.87)	95.5 (5.29)	95.5(6.10)	94.8(1.26)	94.8(1.33)	94.9(1.39)	94.9(1.64)
0.3	0.7	95.5 (5.12)	95.4 (5.38)	95.5 (5.83)	95.5(6.55)	95.1(1.42)	95.0(1.47)	94.9(1.53)	94.9(1.73)
0.5	0.5	95.4 (6.35)	95.4 (6.57)	95.6 (6.84)	95.5(7.38)	95.1(1.73)	95.0(1.75)	94.8(1.79)	95.1(1.96)
0.7	0.3	95.5 (7.48)	95.5 (7.69)	95.6 (7.91)	95.6(8.16)	95.1(2.03)	95.1(2.05)	95.1(2.08)	95.2(2.17)
0.8	0.2	95.5 (8.13)	95.5 (8.23)	95.5 (8.41)	95.5(8.57)	95.1(2.18)	95.0(2.21)	95.2(2.22)	95.3(2.28)
0.95	0.05	95.6 (9.08)	95.5 (9.19)	95.4 (9.14)	95.6(9.24)	95.2(2.41)	95.2(2.43)	95.2(2.43)	95.1(2.44)

*Cell entry is the actual coverage (the average value of the upper confidence limit on the natural log scale is in parentheses). Results based on 50,000 simulations.

For the range of sample sizes evaluated for three sites, the coverage is close to the nominal 95%, and never anti-conservative. For five sites, there is slightly less than 95% coverage at very low between-site variance. The coverage is never less than 94.4%.

The average values of the upper confidence limits tend to increase as the total number of measurements decrease from 9 to 8 to 7. However, with just one day at one site, the configuration (3,3,1) has higher average upper confidence limits than the configuration (3,2,2), although they both have seven days of sampling. Also, as the between-site variance increases the differences in the average upper confidence limits become smaller for these designs. For the 0.95 between-site variance there is almost no difference—all average upper confidence limits are between 9.08 and 9.24 for the three site data. When there is almost no day to day variability, the value of the confidence limit depends only on the between-site variance.

Other simulations have been done for different total variance. Table C-8 has analogous results to those in Table C-7 for total natural log scale variance 0.25 and 3.0, values that were also evaluated in Table C-4 for balanced data.

Table C-8: Coverage (expressed as percentage of simulations) of the modified MOVER method for the unbalanced design based on 3 and 5 sites. 10,000 simulations for each specification of parameter values[+], except that 50,000 were used for the balanced data, as reproduced from Table C-7.

Total Variance	σ^2_s	σ^2_s/Total Variance	Number of days at each site with k=3 sites				Number of days at each site with k=5 sites			
			(3,3,3)	(3,3,2)	(3,2,2)	(3,3,1)	(3,3,3,3,3)	(3,3,3,3,2)	(3,3,3,2,2)	(3,3,2,2,1)
0.25	0	0	95.1(0.90)	95.2(1.04)	95.0(1.14)	95.4(1.41)	94.6(0.33)	94.2(0.35)	94.8(0.38)	94.7(0.46)
	0.05	0.2	95.1(1.22)	94.9(1.33)	95.5(1.42)	95.6(1.63)	94.4(0.41)	94.9(0.43)	94.4(0.44)	94.6(0.51)
	0.075	0.3	95.0(1.37)	94.9(1.43)	95.3(1.56)	95.4(1.73)	94.8(0.46)	94.6(0.47)	94.6(0.48)	94.8(0.54)
	0.125	0.5	95.1(1.67)	95.6(1.72)	95.2(1.82)	95.1(1.94)	94.7(0.54)	94.8(0.55)	94.5(0.56)	94.7(0.60)
	0.175	0.7	95.2(1.99)	95.5(2.02)	95.4(2.08)	95.3(2.16)	94.7(0.62)	94.7(0.63)	94.6(0.64)	94.7(0.66)
	0.2	0.8	95.5(2.14)	95.6(2.19)	95.2(2.19)	94.9(2.24)	94.6(0.66)	95.1(0.67)	94.5(0.66)	94.7(0.68)
	0.238	0.95	95.1(2.38)	94.8(2.35)	95.5(2.36)	95.4(2.41)	94.9(0.72)	94.5(0.72)	94.7(0.72)	94.8(0.73)
3.0	0	0	95.5(9.52)	95.6(11.08)	95.5(12.46)	95.2(15.50)	94.6(2.58)	94.2(2.78)	94.9(3.00)	94.5(3.92)
	0.6	0.2	95.8(13.17)	95.8(14.38)	95.5(15.37)	95.8(18.16)	95.3(3.48)	95.3(3.72)	95.1(3.83)	95.3(4.56)
	0.9	0.3	95.6(14.87)	95.4(16.17)	95.5(17.41)	95.4(19.15)	95.2(3.92)	95.5(4.10)	95.0(4.23)	95.0(4.93)
	1.5	0.5	95.7(18.81)	95.5(19.54)	95.7(20.18)	95.6(21.55)	95.1(4.83)	95.5(4.94)	95.5(5.07)	95.1(5.53)
	2.1	0.7	95.5(22.31)	95.5(22.79)	95.4(23.56)	95.4(24.18)	95.0(5.73)	95.0(5.88)	95.1(5.89)	95.1(6.21)
	2.4	0.8	95.4(24.27)	95.6(24.80)	95.6(24.99)	95.5(25.65)	95.2(6.20)	95.1(6.27)	95.2(6.30)	95.2(6.50)
	2.85	0.95	95.5(26.92)	95.9(26.95)	95.6(27.56)	95.4(27.33)	95.1(6.86)	95.2(6.93)	95.3(7.01)	94.9(6.96)

[+]There are relatively small differences between the simulation results for 10,000 and 50,000 simulations. Also, they are intended mainly to see if there is consistency with the 50,000 simulation results in Tables C-4 and C-7.
*Cell entry is the actual coverage (the average value of the upper confidence limit on the natural log scale is in parentheses).

The results are similar to those in Table C-4 and C-7. For three sites, the coverage is always at least 95%, whereas for 5 sites at total variance 0.25, the coverage is a little less than 95%. The coverage is never less than 94.2%. For three or five sites, the average upper confidence limit increases as the total number of measurements decreases. This should result in some loss of statistical power. As in Table C-7, for the 0.95 ratio of between-site variance to total variance, there is almost no difference in expected values of the upper confidence limits, for the different designs at the given sample size and total variance.

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Appendix D: Use of the Spreadsheet

The Excel spreadsheet "mover_upper_confidence_limits.xlsx" is included with this document. It contains five worksheets: "Ex. Strategy 1-Balanced", "Ex. Strategy 1-Unbalanced", "Ex. Strategy 2", "Blank Strategy 1", and "Blank Strategy 2". The first three of these worksheets have hypothetical example data that were discussed in Appendix B. The two worksheets "Blank Strategy 1" and "Blank Strategy 2" do not have data and should be used by manufacturers to input their data from field testing to calculate results. The user should choose these "blank" sheets according to the sampling strategy used, and enter the data in the red portion of the sheet. The upper confidence limit and other calculated estimates will then appear in bold at the bottom of the sheet on the right side. Whether the data for strategy 1 are balanced or unbalanced, the sheet "Blank Strategy 1" can be used. Whichever blank sheet is selected, that sheet should be copied elsewhere so that the blank sheet is available for new data.

Appendix E: Example Daily Dust Control Checklist

Table provided by Payne & Dolan, Inc.

WATER SPRAY AND EVACUATION CONTROL SYSTEMS DAILY CHECK LIST																	
CHECK LIST COVERS 2 WEEK PERIOD. FILL OUT COMPLETELY DAILY PRIOR TO START-UP. SEND TO _____ WHEN FILLED UP																	
EQUIPMENT #	Fill in Date and Initials, Y = YES = ALL Items Functioning Correctly N = NO = Item not Functioning Correctly.																
WATER SYSTEM																	
DATE																	
	EXAMPLE = 4/11/14																
INSPECTOR																	
	PRINT INITIALS																
WATER PUMP																	
	FUNCTIONING CORRECTLY THIS ITEM CAN POTENTIALLY BE A DEAD LINE ITEM																
PUMP PRESSURE																	
	SET TO MANUFACTURERS. SPEC. VISUALLY READ PRESSURE GAUGE																
INSPECT IN-LINE FILTER																	
	ENSURE IT'S CLEAN, NO DEBRIS																
SPRAY BAR																	
	LOCATION & OPERATION																
NOZZLES																	
	SPRAY / PATTERN / CLEAN																
FLASHING																	
	CONDITION / PLACEMENT AT TRANSFER POINTS																
EVACUATION	ANY OF THE ITEMS UNDER EVACUATION THAT DO NOT OPERATE PROPERLY COULD BE CONSIDERED DEAD LINE ITEMS.																
FAN OPERATES PER																	
MFG.'S SPEC.	FAN SPEED MFG. SPEC. = AIR FLOW - READ HYDRAULIC PRESSURE GAUGE																
	This Function will be periodically checked during the season and tested annually during winter maintenance																
VACUUM TUBES																	
MEET MFG.'S CFM	IF OPERATING CORRECTLY, VACUUM WILL SUCK DUST VALVES SHUT																
	This Function will be periodically checked during the season and tested annually during winter maintenance																
VACUUM TUBES																	
	NO DEBRIS OR CLOGGING IN TUBES AND AIR DUCTS																
	During the day Open Dust Valve by Pressing the Dust Dump Valve Together and Empty.																
FLASHING / SEALING																	
	VISUALLY INSPECT ALL FLASHING TO ENSURE TIGHT SEAL																
EVERY 50 OPERATIONAL HOURS: INSPECT PIPES/HOSES - REMOVE HOSES & REMOVE DUST DEPOSITS, CLEAN OUT CHASSIS AIR DUCTS.																	
COMMENTS:																	

Table provided by Payne & Dolan, Inc.

Appendix F: NAPA/AEM Best Practice Bulletin 1/12

PROJECT PREPARATION

- Are there an adequate number of trained personnel for safe operation?
  - ◊ According to OSHA, each company needs to have a silica program to ensure that all employees have been trained in silica standards.
- Plan how water will be provided on the job site.
  - ◊ Ensure that water truck has enough capacity. Are permits and the proper hose and pumps on the job site?
- Know the work area. Insure work area is adequate to safely maneuver transport vehicle in work zone.
- Consider environmental conditions, such as freezing temperatures.
  - ◊ Possibly locate source for heated water or antifreeze for water tank.
- Ensure that the water filling points are accessible from the front and rear of the machine on narrow lane closures.
- Ensure that there are proper lighting and markers on the water truck for night operation.
- During winter months, make sure that the water system is winterized to ensure proper operations in cold weather and during shutdown overnight.
- If working at night, provide adequate lighting for safe operation. Have preparations been made to perform maintenance and inspections, and safely transfer water?

Safe operators know the work area and any potential hazards.



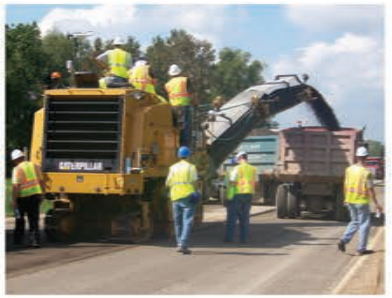
SAFETY RESOURCES



For cold planer Safety Manuals visit:
http://shop.aem.org

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OPERATIONAL GUIDANCE FOR WATER SYSTEMS DURING MILLING OPERATION



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